# Adaptive Demand Allocation Architecture (ADA)

PLC Implementation — Full Replacement

Nebico Oil Storage — Palm Oil Retention & Melting System

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Based on the *VCH Sizing Framework, Second Edition*
*A Unified Design Methodology for Submerged Hydronic Coil Systems*


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This document presents the second architectural revision of the palm oil storage control system, covering trunk and branch sizing, the need-array priority model, P2 governance, load-responsive operating modes, water-side hydraulics, process duty cycles, aggregate demand budgeting, and a fully specified dynamic water-temperature control law. The work was independently researched, modelled, and authored by Sambridda Ranabhat.


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# Contents

# 1 PLC Implementation: Processes and Logic to Insert

This section consolidates every process, register, timer, alarm, I/O point, and control block required to implement ADA on the PLC. Digital I/O is taken directly from the project's own AutoCAD Electrical panel schematic (sheet 6 of 10). Analog I/O tag numbers await the Analog Card sheet; channel *counts* and correspondence are confirmed below. Sections are ordered by dependency (foundational blocks first, blocks that consume their outputs after) rather than by when each piece was originally drafted.

## 1.1 Analog Inputs (AI)

**Table 1.** Analog input channel assignment.

| Channel | Correspondence | Signal | Register |
|---|---|---|---|
| AI-1 | Level – $T_5$ | 4–20 mA (ultrasonic) | $\alpha_S$ |
| AI-2 | Level – TA | 4–20 mA (ultrasonic) | $\alpha_{10A}$ |
| AI-3 | Level – TB | 4–20 mA (ultrasonic) | $\alpha_{10B}$ |
| AI-4 | Level – TC | 4–20 mA (ultrasonic) | $\alpha_{10C}$ |
| AI-5 | Temperature – $T_5$ | 4–20 mA (Pt100 transmitter) | $\theta_S$ |
| AI-6 | Temperature – TA | 4–20 mA (Pt100 transmitter) | $\theta_{10A}$ |
| AI-7 | Temperature – TB | 4–20 mA (Pt100 transmitter) | $\theta_{10B}$ |
| AI-8 | Temperature – TC | 4–20 mA (Pt100 transmitter) | $\theta_{10C}$ |
| AI-9 | Temperature – Solar loop | 4–20 mA (Pt100 transmitter) | $\theta_{\mathrm{Sol}}$ |
| AI-10 | Temperature – Thermal battery | 4–20 mA (Pt100 transmitter) | $\theta_H$ |
| AI-11 | Flow – Trunk/outlet (FLC220) | 4–20 mA, confirmed | Flow rate |

Real channel-to-terminal addressing awaits the Analog Card schematic; AI-1–AI-11 are placeholder labels, not PLC tag names to be programmed as-is.

### 1.1.1 Reading AI Channels

```
FROM K0 K10 D0 K1        // read raw ADC count (K0/K10 = PLACEHOLDER
                          // unit number / buffer address)
MUL  D0  K100  D2         // D2 = raw x range_max (PLACEHOLDER K100)
DIV  D2  K4000 D4         // D4 = engineering value (PLACEHOLDER K4000
                          // = ADC full-scale count)
```

Every constant above is a placeholder: unit number and buffer address depend on the installed module; `K4000` assumes a 12-bit-class module; `K100` is a worked-example range only, replaced per channel; whether raw count 0 corresponds to 4 mA or 0 mA must be confirmed per channel or every reading carries a silent offset error.

## 1.2 Digital Inputs (DI)

**Table 2.** Digital input tags, panel schematic sheet 6/10.

| Tag | Correspondence | Role |
|---|---|---|
| X0 | E-STOP | Master safety |
| X1 | Automation Enable/Disable | Master mode select |
| X2 | Circulation Pump (manual) | VFD-01 / trunk manual button |
| X3 | Oil Distribution Pump (manual) | VFD-02 manual button |
| X4 | Jacket Pump (manual) | Jacketed-pipe circulation manual button |
| X5 | AODD Pump (manual) | $T_{10} \rightarrow T_5$ transfer manual button |
| X6 | Float – Heating tank (HEA), NC | Low-level fault input |
| X7 | Float – Solar tank (SOL), NC | Low-level fault input |
| X10 | Loop Exit | Manual-mode solar/bypass override |

### 1.2.1 X10 – Loop Exit (Manual-Mode Only)

```
IF Automation Disabled (X1=OFF):
    X10=ON  -> close SV104 (Y15), SV105 open -> through-solar; Y25=ON
    X10=OFF -> close SV105 (Y16), SV104 open -> bypass;       Y25=OFF
IF Automation Enabled (X1=ON):
    Automatic theta_Sol vs T_water comparison governs
    (Section~\ref{sec:plc-solaraux})
```

### 1.2.2 X6/X7 – Float Switches (NC, Warning Only)

```
X6=OFF (contact open) -> HMI warning: "HEA water loop level low"
X7=OFF (contact open) -> HMI warning: "SOL water loop level low"
```

Deliberately warning-only — no pump interlock, no forced bypass. Automated reservoir top-up is deferred future work pending clearance. Confirm NC polarity against the actual float switch datasheet before commissioning.

## 1.3 Digital Outputs (DO)

**Table 3.** Digital output tags, panel schematic sheet 6/10.

| Tag | Correspondence | Role |
|---|---|---|
| Y0 | E-STOPPED | Master safety status lamp |
| Y1 | Automation Enabled | Master mode status lamp |
| Y2 | Circulation Pump (HSM) | Trunk: 2-destination speed select |
| Y3 | Oil Distribution Pump (Active) | VFD-02 status lamp |
| Y4 | Jacket Pump (Active) | Status lamp |
| Y5 | AODD Pump (Active) | Status lamp |
| Y6 | Circulation Pump (LSM) | Trunk: 1-destination speed select |
| Y7 | Heating Element 1 | Auxiliary immersion element (12 kW) |
| Y10 | Heating Element 2 | Auxiliary immersion element (12 kW) |
| Y11 | SV100 | $T_5$ branch valve |
| Y12 | SV101 | TA branch valve |
| Y13 | SV102 | TB branch valve |
| Y14 | SV103 | TC branch valve |
| Y15 | SV104 | Solar selector – through-solar |
| Y16 | SV105 | Solar selector – bypass-solar |
| Y17 | Circulation Pump (FSM) | Trunk: 3-destination speed select |
| Y20 | Heating Element 3 | Auxiliary immersion element (12 kW) |
| Y21 | PBV100 | $T_5$ oil valve |
| Y22 | PBV101 | TA oil valve |
| Y23 | PBV102 | TB oil valve |
| Y24 | PBV103 | TC oil valve |
| Y25 | Solar Cycle Active | Status lamp |

Y7/Y10/Y20 confirm the auxiliary array at 3 × 12 kW (36 kW), matching the peak-wattage analysis basis without adjustment. Y20 remains Heating Element 3; Y25 is the separate, previously-conflicting "solar active" indicator, now on its own tag.

## 1.4 Master Safety and Mode Control: Ladder Logic

**Safety-engineering caveat.** The logic below assumes E-STOP is a **hardwired safety relay/contactor circuit** cutting power directly, independent of the PLC scan cycle, with X0 wired as a status/monitoring input only. Relying on PLC-scanned logic as the sole emergency-stop mechanism is not standard safe machinery practice (IEC 60204-1). Confirm a hardwired circuit exists upstream of X0.

```
LDI X0
SET M1_ESTOP            // latches on X0 going OFF

LD  X_EStopReset          // PLACEHOLDER -- reset input not yet defined
RST M1_ESTOP
```

```
LD  M1_ESTOP
OUT Y0

LDI M1_ESTOP
OUT M2_SystemOK          // ANDed into every output rung, manual+auto

LD  X1
OUT Y1
LD  X1
OUT M3_AutoEnabled
```

Open items: (1) what clears the E-STOP latch — no dedicated reset input defined; (2) behaviour on a mid-operation switch from Automation Enabled to Disabled — abrupt stop vs. complete-current-step, not yet decided.

### 1.4.1 Gating Pattern: Manual vs. Automatic

```
// Manual -- live only when System OK AND Automation Disabled
LD  M2_SystemOK
ANI M3_AutoEnabled
AND X2                    // e.g. Circulation Pump manual button
OUT [pump output]

// Automatic -- live only when System OK AND Automation Enabled
LD  M2_SystemOK
AND M3_AutoEnabled
AND [arbiter / need-array condition]
OUT [pump output]
```

This pattern repeats for every pump and every valve/gating block in this document — `M2_SystemOK` and `M3_AutoEnabled` are the two conditions every real-world output coil ultimately traces back to.

## 1.5 PO Tank Designation: HMI Selector and Ladder Logic

**Register convention:** 0=None, 1=TA, 2=TB, 3=TC. Designation is optional — 0 is a valid standing state. SV/PBV-to-tank assignment is fixed at commissioning; designation only changes which *logical* process (PO maintenance vs. palm oil melt) a tank's existing valve serves, not the valve wiring itself.

```
LD  X_Confirm
PLS M100                  // one-scan pulse on rising edge only

LD  M100
MOV D50  D51                // commit: pending -> active
```

```
LD= D51 K0   OUT M0         // no PO tank
LD= D51 K1   OUT M1         // TA is PO
LD= D51 K2   OUT M2         // TB is PO
LD= D51 K3   OUT M3         // TC is PO
```

No automated contamination interlock, by explicit decision (budget constraint) — operator's manual valve is the sole safeguard. Access level on `X_Confirm` (standard tap vs. supervisor PIN) remains an open item.

## 1.6 Calculated Registers: Ladder Logic

Fixed-point arithmetic throughout (native float support unconfirmed). Percentages as plain integers, kW as integers, temperatures in tenths of a degree unless stated otherwise.

### 1.6.1 10-Second Refresh Oscillator

```
LDI T0
OUT T0 K100              // 10.0s
LD  T0
OUT M10_Refresh           // one-scan pulse, every 10s
```

### 1.6.2 $T_{\text{water}}$ Target

Implements $T_{\text{water}} = 40 + 4.87P$ [kW], clamped 40–70 °C:

```
LD  M10_Refresh
MUL D_Ptotal K300  D200   // D200 = P[W] x 300
DIV D200     K6160 D202   // D202 = increment, tenths degree
ADD D202     K400  D204   // D204 = T_water x10, unclamped

LD< D204 K400
MOV K400 D204
LD> D204 K700
MOV K700 D204
MOV D204 D_Twater_target
```

### 1.6.3 Mode Register (Eco/Efficiency/Standard/Full Throttle)

`D_Mode`: 0=Eco, 1=Efficiency, 2=Standard, 3=Full Throttle. Current-state-gated transitions — structurally prevents chattering or state-skipping:

```
LD= D_Mode K0   AND< D_AlphaS K85   MOV K1 D_Mode   // Eco->Eff, 90/85
LD= D_Mode K1   AND>= D_AlphaS K90  MOV K0 D_Mode
LD= D_Mode K1   AND< D_AlphaS K65   MOV K2 D_Mode   // Eff->Std, 70/65
LD= D_Mode K2   AND>= D_AlphaS K70  MOV K1 D_Mode
LD= D_Mode K2   AND< D_AlphaS K45   MOV K3 D_Mode   // Std->FT, 50/45
LD= D_Mode K3   AND>= D_AlphaS K50  MOV K2 D_Mode
```

Open item: power-up default for D_Mode not yet specified.

#### 1.6.4 Suspicion Score ($S \in \{0, 1, 2, 3\}$)

**5-minute** no-response auto-increment (corrected; earlier draft carried a 10-minute figure in one summary table that contradicted the ladder itself):

```
LD  M_Overrun_Pulse         // from P2 block, Section~\ref{sec:p2block}
INC D_Suspicion

LD> D_Suspicion K3
MOV K3 D_Suspicion           // cap at 3

LD= D_Suspicion K1  OUT M_HMIPrompt
LD= D_Suspicion K2  OUT M_Alarm
LD= D_Suspicion K3  OUT M_DisableFeed
LD= D_Suspicion K3  OUT M_Alarm

LD  M_PromptActive
OUT T5  K3000                // 5 min = 300s
LD  T5
INC D_Suspicion
RST M_PromptActive

LD  M_DailyReset_Pulse       // PLACEHOLDER -- RTC/day-rollover
MOV K0 D_Suspicion
```

#### 1.6.5 Auxiliary Stage Register (0/1/2/3)

Drives HE1/HE2/HE3 (Y7/Y10/Y20) staging against $E_{\text{aux,required}} = E_{\text{total}}$ (evaluated only while bypass-solar is the active driver). Thresholds (26/22, 14/10, 0/2 kW, ±2 kW hysteresis) are proposed, not independently confirmed.

```
LD= D_AuxStage K0  AND> D_Ereq_kW K0   MOV K1 D_AuxStage
LD= D_AuxStage K1  AND<= D_Ereq_kW K2  MOV K0 D_AuxStage
LD= D_AuxStage K1  AND>= D_Ereq_kW K14 MOV K2 D_AuxStage
LD= D_AuxStage K2  AND< D_Ereq_kW K10  MOV K1 D_AuxStage
LD= D_AuxStage K2  AND>= D_Ereq_kW K26 MOV K3 D_AuxStage
LD= D_AuxStage K3  AND< D_Ereq_kW K22  MOV K2 D_AuxStage

LD>= D_AuxStage K1  OUT Y7    // HE1
LD>= D_AuxStage K2  OUT Y10   // HE2
LD>= D_AuxStage K3  OUT Y20   // HE3
```

#### 1.6.6 Tank Scoring: $S_{\text{casual}}$, $S_{\text{immediate}}$

$S_{\text{casual}} = 0.6\alpha + 0.4\theta_{\text{norm}}$, $S_{\text{immediate}} = 0.4\alpha + 0.6\theta_{\text{norm}}$, per the base VCH framework. $\theta_{\text{norm}}$ (0–100 scale) assumed normalised against the 70 °C water ceiling — this normalisation basis is inferred, not independently confirmed elsewhere. Scores kept ×10 for one-decimal precision.

```
// Per tank (worked for TA; repeat identically for TB, TC)
MUL D_ThetaTA K100 D410
DIV D410      K700 D_TnormTA        // normalized theta, 0-100

MUL D_AlphaTA K6 D412
MUL D_TnormTA K4 D414
ADD D412 D414 D_ScoreCasualTA        // S_casual x10

MUL D_AlphaTA K4 D416
MUL D_TnormTA K6 D418
ADD D416 D418 D_ScoreImmTA           // S_immediate x10
```

**PO exclusion + ranking (drives $D_{\mathbf{RankedTankTemp}}$, consumed by P1).**

```
// Sentinel-exclude the PO-designated tank
LD  M1  MOV K-1 D_EligCasualTA
LDI M1  MOV D_ScoreCasualTA D_EligCasualTA
LD  M2  MOV K-1 D_EligCasualTB
LDI M2  MOV D_ScoreCasualTB D_EligCasualTB
LD  M3  MOV K-1 D_EligCasualTC
LDI M3  MOV D_ScoreCasualTC D_EligCasualTC

// Max-of-3 with ID tracking (safe pattern: flag first, then apply)
MOV K1 D_MaxID
MOV D_EligCasualTA D_MaxScore

LD> D_EligCasualTB D_MaxScore
OUT M_Challenger
LD M_Challenger  MOV D_EligCasualTB D_MaxScore
LD M_Challenger  MOV K2 D_MaxID

LD> D_EligCasualTC D_MaxScore
OUT M_Challenger
LD M_Challenger  MOV D_EligCasualTC D_MaxScore
LD M_Challenger  MOV K3 D_MaxID

LD= D_MaxID K1  MOV D_ThetaTA D_RankedTankTemp
LD= D_MaxID K2  MOV D_ThetaTB D_RankedTankTemp
LD= D_MaxID K3  MOV D_ThetaTC D_RankedTankTemp
```

Tie-break on equal scores defaults to whichever rung executes first (effectively TA-favoured); not an explicit design decision, flagged for confirmation.

### 1.6.7 Height-to-Mass Conversion

Flat-surface assumption (cylindrical tanks, level treated as flat, per explicit instruction). Density $\rho = 957.0\,\mathrm{kg/m^3}$, back-derived precisely from the base document's own

8900 kg/9.300 m$^3$ full-tank figures — **correcting an earlier, less careful ≈890 kg/m$^3$ estimate used in intermediate chat discussion.**

**Table 4.** Height-to-mass constants.

| Tank | $D$ (m) | Mass/mm | Max height |
|---|---|---|---|
| $T_5$ | 1.80 | 2.4353 kg/mm | 2000 mm |
| $T_{10}$ (TA/TB/TC) | 2.32 | 4.0455 kg/mm | 2200 mm |

```
// T5 (worked example; T10 tanks use K40455 and K2200 instead)
SUB K2000 D_DistT5 D_HeightT5

DMUL D_HeightT5 K24353 D510    // = mass(kg) x 10000, 32-bit
DDIV D510       K10000 D_MassT5 // whole kg, fits 16-bit (max ~4871)
```

## 1.7 Thermal Blending Register: $T_f$ Substitution

$$m_w = \rho_w \cdot \frac{\pi}{4} D_i^2 L \approx 14.34\,\text{kg}$$
$$m_{ss} = \rho_{ss} \cdot \frac{\pi}{4} (D_o^2 - D_i^2) L \approx 31.66\,\text{kg}$$
$$A \equiv m_w c_w + m_{ss} c_{ss} \approx 75\,808\,\text{J/K}$$

$T_f(\theta_H, \theta_S, m_p) = \dfrac{75808\,\theta_H + 2000\,m_p\,\theta_S}{75808 + 2000\,m_p}$ — $x = \theta_H$ (thermal buffer), $y = \theta_S$ ($T_5$), **confirmed** against the base document's original $T_f(x, y, m_p)$ definition ("$x$ = buffer tank temperature, $y$ = target melting tank temperature").

Schedule 10S wall thickness assumed for the 1.5" coil; cross-check if the as-built coil uses a different schedule. $A$ exceeds 16-bit range — every step below uses 32-bit (`DMUL`/`DADD`/`DDIV`), a deliberate departure from the rest of Section 1.6.

```
MOV D_ThetaH D330   MOV K0 D331     // zero-extend to 32-bit
MOV D_ThetaS D332   MOV K0 D333
MOV D_Mp     D334   MOV K0 D335

DMUL K75808 D330 D340               // A x theta_H
DMUL D334   K2000 D344               // m_p x c_p
DMUL D344   D332  D350               // (m_p*c_p) x theta_S

DADD D340   D350  D360               // numerator
DADD K75808 D344  D362               // denominator

DDIV D360 D362 D370
```

```
MOV  D370 D_Tf

LD>= D_Tf K550                   // 55.0 degC x10
OUT  M_TfInterlock
LD   M_TfInterlock
OUT  M4_ThrottleOrDivert         // feeds P1 below
```

Open items: (1) $m_p$ has no defined source register or acquisition method — operator-entered, level-derived, or fixed-constant, not yet decided; without it `D_Mp` is inert. (2) Runs every scan (unconditional), per P1's "continuously evaluate" wording — a deliberate deviation from the 10s pattern elsewhere, given this is a safety interlock. (3) `DDIV` truncates toward zero — $T_f$ can read up to 0.09 °C below true value, the non-conservative direction for a $\geq$ trip; likely immaterial but should be an informed decision.

### 1.8 P1 — $T_5$ Block

```
LD<  D_ThetaS K400
OUT  M_P1Demand
LD   M_P1Demand
OUT  Y11                      // SV100

LD   M_TfInterlock
OUT  M4_ThrottleOrDivert      // Section~\ref{sec:tf}

LD<   D_AlphaS K50
AND>= D_RankedTankTemp K360   // Section~\ref{sec:scoring-ladder}
OUT   M_BatchRelease
LD    M_BatchRelease
OUT   Y5                      // AODD Pump
```

### 1.9 P2 — PO Tank Block

**No pulse-hold timer in this revision.** The original 20s-on/60s design existed to accommodate branch scarcity under the two-branch architecture. Under the 3-wide need array, PO demand can simply be served normally by the arbiter for as long as it remains live — there is no scarcity constraint left to justify throttling. Dispense stops and the demand flag clears on operator confirmation, not on a duty-cycled hold.

```
LD  X_Confirm_Dispense          // HMI: n_d entered, dispense started
// [dispense-duration calc from n_d -- volumetric, pump-rate dependent,
//  not detailed here; drives the dispense valve open/close directly]

LD  D_DispensedVol
AND>= D_DispensedVol D_2xNd      // actual draw >= 2 x n_d
OUT M_Overrun_Pulse               // consumed by Suspicion ladder,
                                  // Section~\ref{sec:registers-ladder}
```

```
LD  M1  OR M2  OR M3              // any PO tank designated
AND [dispense active, not yet "process complete"]
OUT M_P2Demand
```

## 1.10 Need-Array Arbiter: Generalised Four-Destination Model

**Four possible destinations** ($T_5$, TA, TB, TC), **trunk capacity for three at once.** LSM/HSM/FSM correspond directly to 1/2/3 simultaneously-served destinations, matching the confirmed flow-logic table rows 1–4 (LSM), 5–10 (HSM), 11–14 (FSM). $T_5$ is always guaranteed a slot when demanding; the PO tank (if designated and demanding) is guaranteed the next slot; the remaining slot(s) go to $T_{10}$ tanks ranked by $S_{\text{immediate}}$ (Section 1.6.6).

```
// Raw per-destination demand
LD  M_P1Demand              OUT M_D_T5    // Section~\ref{sec:p1block}
LD  M1  AND M_P2Demand      OUT M_D_TA_asPO
LD  M2  AND M_P2Demand      OUT M_D_TB_asPO
LD  M3  AND M_P2Demand      OUT M_D_TC_asPO

LDI M1  AND< D_ThetaTA K360  OUT M_D_TA_asMelt   // not PO, below 36C
LDI M2  AND< D_ThetaTB K360  OUT M_D_TB_asMelt
LDI M3  AND< D_ThetaTC K360  OUT M_D_TC_asMelt

LD M_D_TA_asPO  OR M_D_TA_asMelt  OUT M_D_TA
LD M_D_TB_asPO  OR M_D_TB_asMelt  OUT M_D_TB
LD M_D_TC_asPO  OR M_D_TC_asMelt  OUT M_D_TC

MOV K0 D_NeedCount
LD M_D_T5  INC D_NeedCount
LD M_D_TA  INC D_NeedCount
LD M_D_TB  INC D_NeedCount
LD M_D_TC  INC D_NeedCount

// Default: serve everyone demanding (correct for count 0-3)
LD M_D_T5  OUT M_Serve_T5
LD M_D_TA  OUT M_Serve_TA
LD M_D_TB  OUT M_Serve_TB
LD M_D_TC  OUT M_Serve_TC

// 4-way tie-break, PO exists: drop lower-S_immediate of the OTHER two
LD= D_NeedCount K4  ANI M0  AND M1
LD< D_ScoreImmTB D_ScoreImmTC   RST M_Serve_TB
LD= D_NeedCount K4  ANI M0  AND M1
LD>= D_ScoreImmTB D_ScoreImmTC  RST M_Serve_TC

LD= D_NeedCount K4  ANI M0  AND M2
LD< D_ScoreImmTA D_ScoreImmTC   RST M_Serve_TA
```

```
LD= D_NeedCount K4  ANI M0  AND M2
LD>= D_ScoreImmTA D_ScoreImmTC  RST M_Serve_TC

LD= D_NeedCount K4  ANI M0  AND M3
LD< D_ScoreImmTA D_ScoreImmTB   RST M_Serve_TA
LD= D_NeedCount K4  ANI M0  AND M3
LD>= D_ScoreImmTA D_ScoreImmTB  RST M_Serve_TB

// 4-way tie-break, no PO tank: drop lowest of all three
LD= D_NeedCount K4  AND M0
AND< D_ScoreImmTA D_ScoreImmTB  AND< D_ScoreImmTA D_ScoreImmTC
RST M_Serve_TA
LD= D_NeedCount K4  AND M0
AND< D_ScoreImmTB D_ScoreImmTA  AND< D_ScoreImmTB D_ScoreImmTC
RST M_Serve_TB
LD= D_NeedCount K4  AND M0
AND< D_ScoreImmTC D_ScoreImmTA  AND< D_ScoreImmTC D_ScoreImmTB
RST M_Serve_TC

// Serve count -> trunk state
MOV K0 D_ServeCount
LD M_Serve_T5  INC D_ServeCount
LD M_Serve_TA  INC D_ServeCount
LD M_Serve_TB  INC D_ServeCount
LD M_Serve_TC  INC D_ServeCount

LD M2_SystemOK AND M3_AutoEnabled AND= D_ServeCount K1  OUT Y6   // LSM
LD M2_SystemOK AND M3_AutoEnabled AND= D_ServeCount K2  OUT Y2   // HSM
LD M2_SystemOK AND M3_AutoEnabled AND= D_ServeCount K3  OUT Y17  // FSM

// Valve drive
LD M_Serve_T5  OUT Y11    LD M_Serve_TA  OUT Y12
LD M_Serve_TB  OUT Y13    LD M_Serve_TC  OUT Y14
```

Tie-break on equal $S_{\text{immediate}}$ scores defaults to whichever comparison rung executes first (TA-favoured); consistent with the same open item flagged in Section 1.6.6.

### 1.10.1 Manual Path

```
LD  M2_SystemOK
ANI M3_AutoEnabled
AND X2
OUT Y2                     // manual: fixed HSM only, no LSM/FSM select
```

Open item: no manual LSM/FSM selection exists on the physical panel (single trunk button). Confirm acceptable or add a speed-select input.

### 1.11 Mode State Machine

Fully laddered in Section 1.6; not repeated here to avoid two copies drifting apart. Independent of, and not to be confused with, the trunk's LSM/HSM/FSM state (Section 1.10).

### 1.12 Pump Control Blocks

**Table 5.** Pump I/O summary.

| Pump | Manual (X) | Status/Select (Y) | Auto-trigger |
|---|---|---|---|
| Circulation (VFD-01) | X2 | Y6/Y2/Y17 | Need-array count (Section 1.10) |
| Oil Distribution (VFD-02) | X3 | Y3 | Any of $\delta_X, \delta_Y, \delta_Z$ live |
| Jacket | X4 | Y4 | Not modelled — confirmed I/O point only |
| AODD (HPP) | X5 | Y5 | P1 batch-release (Section 1.8) |

### 1.13 Solar and Auxiliary Gating Block

```
LD  M3_AutoEnabled
AND> D_ThetaSol D_Twater_target
OUT Y15                        // through-solar, automatic

LDI M3_AutoEnabled
AND X10
OUT Y15                        // through-solar, manual (Loop Exit)

LD  M3_AutoEnabled
AND<= D_ThetaSol D_Twater_target
OUT Y16                         // bypass, automatic

LDI M3_AutoEnabled
ANI X10
OUT Y16                         // bypass, manual

LD  Y15
OUT Y25                         // lamp mirrors actual valve state
```

Auxiliary staging (HE1/HE2/HE3) is fully laddered in Section 1.6 — not repeated here.

## 1.14 Oil Distribution Block

Priority X > Y > Z, per the base VCH framework. Valve combinations per the confirmed flow-logic table (taken as given, not re-derived):

```
LD  M_DemandX
OUT M_ServeX

LDI M_DemandX  AND M_DemandY
OUT M_ServeY

LDI M_DemandX  ANI M_DemandY  AND M_DemandZ
OUT M_ServeZ

LD M_ServeX  OUT Y22            // close PBV101 (TA) -> port x
LD M_ServeY  OUT Y21            // close PBV100 (T5)
LD M_ServeY  OUT Y24            // close PBV103 (TC) -> port y
LD M_ServeZ  OUT Y21            // close PBV100 (T5)
LD M_ServeZ  OUT Y23            // close PBV102 (TB) -> port z

LD M_ServeX  OR M_ServeY  OR M_ServeZ
OUT Y3                           // Oil Distribution Pump active
```

## 1.15 Fault Detection Block

### 1.15.1 Temperature Rate-of-Change (Physics-Based)

Max physically achievable rate computed live from current mass, not a fixed constant — a tank at low fill has a different thermal-mass-driven ceiling than one nearly full.

```
// Worked for TA; repeat for TB, TC, T5 with that tank's own constants
LD  M10_Refresh
SUB D_ThetaTA D_ThetaTA_Prev D_DeltaThetaTA
LD  M10_Refresh
MOV D_ThetaTA D_ThetaTA_Prev

// Max heating rate this cycle (tenths degC per 10s), live mass
DMUL K5011  K100  D600          // Q_max(W) x 100, single/dual coil rate
DDIV D600   D_MassTA D602        // / mass(kg)
DDIV D602   K2000   D_MaxHeatRateTA  // / Cp -> tenths degC/10s (scaled)

// Max cooling rate, using tank's own vessel-loss U*A (live ThetaTA)
SUB  D_ThetaTA K181 D610         // (theta - ambient 18.1C x10)
DMUL K668  D610  D620             // U_cyl*A_cyl+U_base*A_base
                                  // (T10, W/K x10) x deltaT
DDIV D620  D_MassTA D622
DDIV D622  K2000    D_MaxCoolRateTA

LD>  D_DeltaThetaTA D_MaxHeatRateTA
```

```
OUT  M_FaultHeatingTA
LD<  D_DeltaThetaTA K0
AND> D_DeltaThetaTA D_MaxCoolRateTA   // magnitude check on cooling side
OUT  M_FaultCoolingTA
```

`K5011` (T10 single/dual coil rate, W) and `K668` (combined vessel-loss coefficient, W/K ×10) are pulled from figures already established elsewhere in this document series; confirm against the specific tank/duty before finalising, since $T_5$ and the PO-tank duty use different constants (their own established $\varepsilon C_w$ and loss coefficients).

### 1.15.2 Level/Mass Rate-of-Change (Activity-Gated)

Simpler and more robust than a rate threshold: outside a known active transfer window, *any* mass change is anomalous.

```
LD  M10_Refresh
SUB D_MassTA D_MassTA_Prev D_DeltaMassTA
LD  M10_Refresh
MOV D_MassTA D_MassTA_Prev

LD  M_ServeTA_AODD           // Section~\ref{sec:p1block} -- active transfer
OR  M_ServeX  OR M_ServeY  OR M_ServeZ   // active dispense (if TA is PO)
OUT M_TAActiveTransfer

LDI M_TAActiveTransfer
AND<> D_DeltaMassTA K0        // any nonzero change while inactive
OUT M_FaultLevelTA
```

### 1.15.3 $T_{10}$ High-Limit Hard Alarm

```
LD>  D_ThetaTA K750           // 75.0 degC x10 (repeat for TB, TC)
OUT  M_HighLimitTA

LD   M_HighLimitTA
OR   M_HighLimitTB
OR   M_HighLimitTC
OUT  M_Alarm
RST  Y7    // cut HE1
RST  Y10   // cut HE2
RST  Y20   // cut HE3
```

## 1.16 Timers

- Suspicion meter: 24 hr reset; **5-min** no-HMI-response auto-increment (corrected).
- $E_{\text{total}}/T_{\text{water}}$/scoring/fault-check refresh: 10 s (`M10_Refresh`).
- Thermal recharge sequence: circulate-to-36 °C, then full-auxiliary.

### 1.17 Alarms and Interlocks

- E-STOP (X0) $\rightarrow$ global output kill, Y0 latched.
- $T_f \geq 55\,^\circ\mathrm{C} \rightarrow \mathrm{T}_5$ throttle/divert.
- $\theta_H < 60\,^\circ\mathrm{C} \rightarrow$ recharge alarm.
- $\mathrm{T}_{10} > 75\,^\circ\mathrm{C} \rightarrow$ alarm + HE1/HE2/HE3 cutoff.
- Temperature rate anomaly (heating or cooling) $\rightarrow$ alarm.
- Level/mass rate anomaly (outside active transfer) $\rightarrow$ alarm.
- Suspicion meter $S$=2 (alarm), $S$=3 (disable + alarm).

PO tank valve/designation-mismatch alarm removed — most valves are now manual, no feedback tag exists to check against.

### 1.18 Two Operator Interfaces

- **Physical panel:** E-STOP, Automation Enable/Disable, 4 pump manual buttons + status lamps, Loop Exit (X0–X10, Y0–Y6, Y25).
- **Touchscreen HMI:** PO tank designation, dispense entry, suspicion meter, mode indicator, fault stop/continue, I/O diagnostic screen, trend logging.

## 2 Summary Ladder: Consolidated Register and Tag Map

All register addresses (D, T, M ranges) are illustrative placeholders throughout this document, chosen only to avoid obvious collisions with each other — final addressing must be coordinated as one complete map before programming, not assembled from these per-section choices as-is.

**Table 6.** Digital I/O — physical tags (confirmed).

| Range | Contents |
|---|---|
| X0–X10 | E-STOP, Automation, 4 manual pump buttons, 2 float switches, Loop Exit |
| Y0–Y25 | Status lamps, VFD speed selects, SV100–105, PBV100–103, HE1–3, solar lamp |

**Table 7.** Working flags (M) by function.

| Group | Flags |
|---|---|
| Safety/mode | M1_ESTOP, M2_SystemOK, M3_AutoEnabled |
| PO designation | M0–M3 (designation state) |
| Demand (raw) | M_P1Demand, M_P2Demand, M_D_T5/TA/TB/TC |
| Arbiter serve | M_Serve_T5/TA/TB/TC, M_Challenger |
| $T_f$/P1 | M_TfInterlock, M4_ThrottleOrDivert, M_BatchRelease |
| P2 | M_Overrun_Pulse, M_PromptActive, M_HMIPrompt, M_Alarm, M_DisableFeed |
| Solar | (none dedicated — drives Y15/Y16/Y25 directly) |
| Oil distribution | M_DemandX/Y/Z, M_ServeX/Y/Z |
| Fault detection | M_FaultHeatingXX, M_FaultCoolingXX, M_FaultLevelXX, M_HighLimitXX, M_TAActiveTransfer |

**Table 8.** Data registers (D) by function.

| Group | Registers |
|---|---|
| AI scaling | D0, D2, D4 (example channel) |
| PO designation | D50 (pending), D51 (active) |
| $T_{\mathrm{water}}$/refresh | D200–D204, D_Ptotal, D_Twater_target |
| Mode | D_Mode, D_AlphaS |
| Suspicion | D_Suspicion |
| Aux staging | D_AuxStage, D_Ereq_kW |
| Scoring | D410–D418, D_ScoreCasualXX, D_ScoreImmXX, D_EligCasualXX, D_MaxScore, D_MaxID, D_RankedTankTemp |
| Mass conversion | D510, D520, D_MassT5, D_MassTA/TB/TC, D_HeightT5/TA/TB/TC |
| $T_f$ | D330–D371, D_ThetaH, D_ThetaS, D_Mp, D_Tf |
| Arbiter | D_NeedCount, D_ServeCount |
| Fault detection | D600–D622, D_DeltaThetaXX, D_MaxHeatRateXX, D_MaxCoolRateXX, D_DeltaMassXX |

**Table 9.** Timers (T).

| Timer | Purpose |
|---|---|
| T0 | 10 s refresh oscillator (self-resetting) |
| T5 | Suspicion meter 5-min no-response auto-increment |

**Every major block in this document, at a glance:** AI scaling → safety/mode gating → PO designation → dynamic $T_{\mathrm{water}}$, mode ladder, suspicion meter, auxiliary staging, tank scoring, height-to-mass → $T_f$ interlock → P1/P2 demand blocks → need-array arbiter → pump/valve outputs, solar gating, oil distribution → fault detection → alarms. Each block's outputs are consumed by name in the block(s) after it, matching the dependency order this document is now arranged in.

## 3 System Estimation and Conclusions

1. **The need-array arbiter is now a genuine four-destination model**, not the earlier two-branch-dedicated approximation — every physical destination ($T_5$, TA, TB, TC) competes on equal footing, with $T_5$ and a designated, demanding PO tank guaranteed their slots and $S_{\text{immediate}}$ resolving the remainder under true four-way contention.
2. **P2's pulse-hold timer is retired, not replaced** — the 3-wide array removes the scarcity condition it existed to manage.
3. **Fault detection is simpler than the original physics-based concept but grounded in the same live-mass, live-rate philosophy** — temperature and level anomalies are both detected relative to what the system's own known capacity and current state make physically plausible, not fixed thresholds tuned to one operating point.
4. **Every open item flagged during development remains genuinely open** (E-STOP reset source, mode power-up default, $m_p$ acquisition method, tie-break on equal scores, manual-mode LSM/FSM access, aux staging hysteresis confirmation) — listed together in the Summary Ladder tables above so none are lost before final register addressing is locked.

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